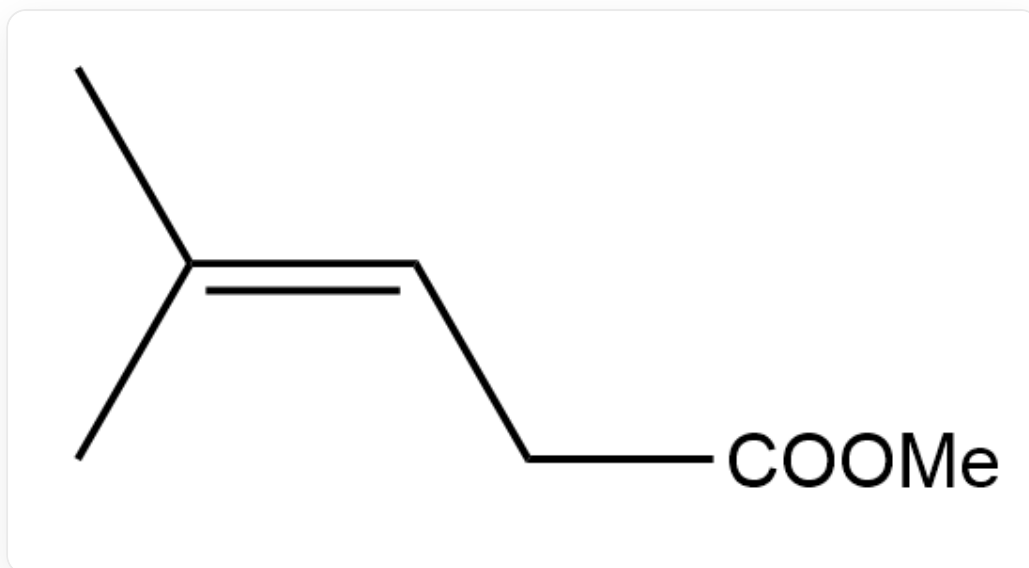


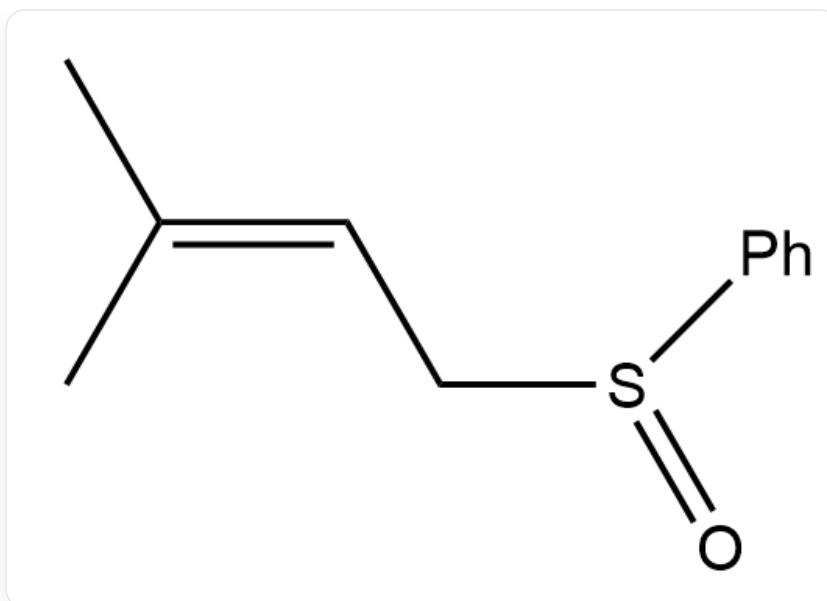
Question

Singlet oxygen 1O_2 can react with the following two substrates. Please provide the main product of each addition reaction (without considering stereoisomers).



C/C(C)=C/CC(OC)=O, Substrate 1

Substrate 1

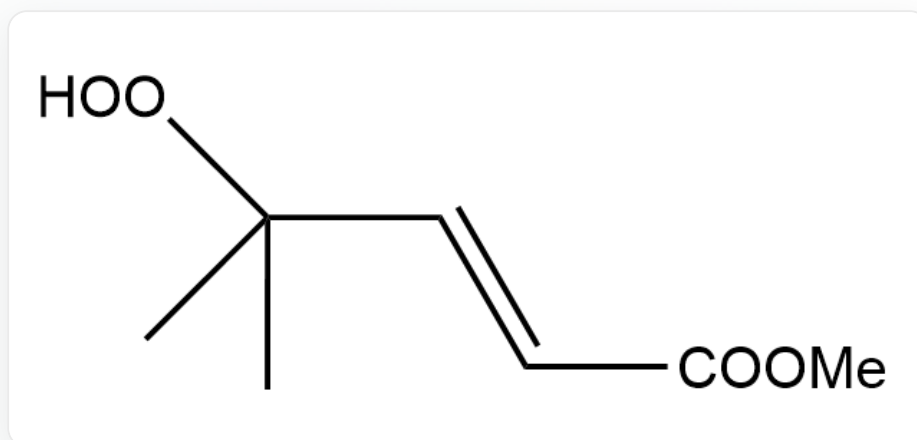


C/C(C)=C/CS(C1=CC=CC=C1)=O, Substrate 2

Substrate 2

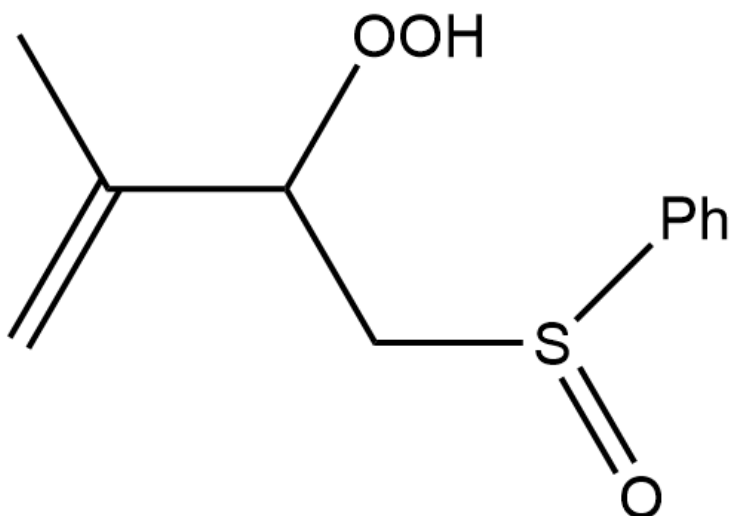
A. All other options are incorrect

B.



CC(C)(OO)/C=C/C(OC)=O, Product 1

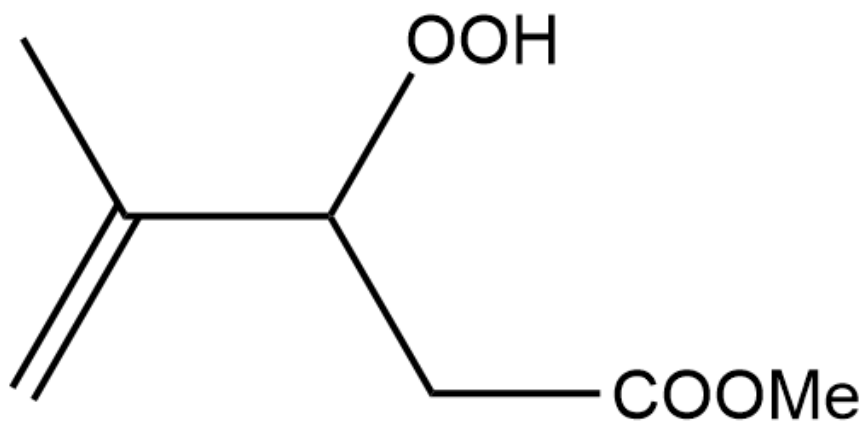
Product 1



CC(C(OO)CS(C1=CC=CC=C1)=O)=C, Product 2

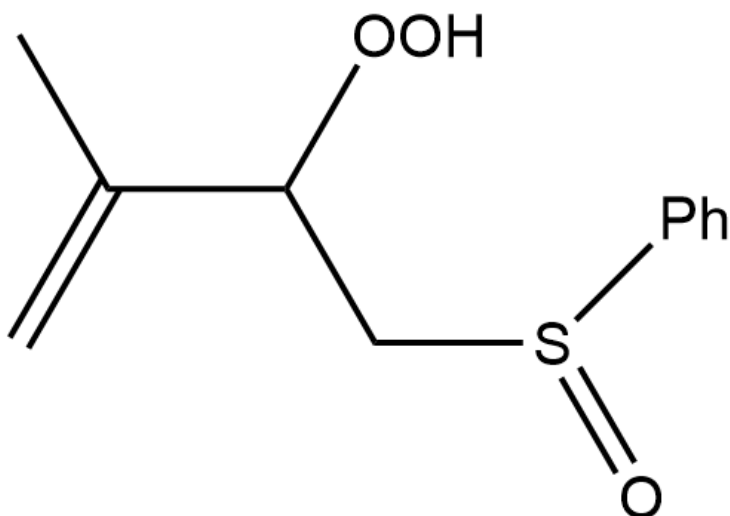
Product 2

C.



CC(C(OO)CC(OC)=O)=C, Product 1

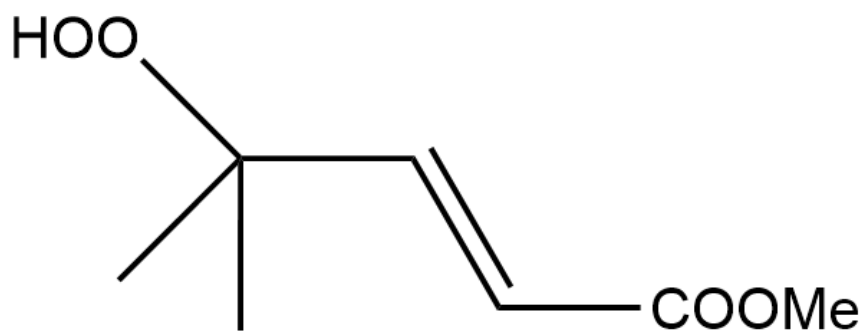
Product 1



CC(C(OO)CS(C1=CC=CC=C1)=O)=C, Product 2

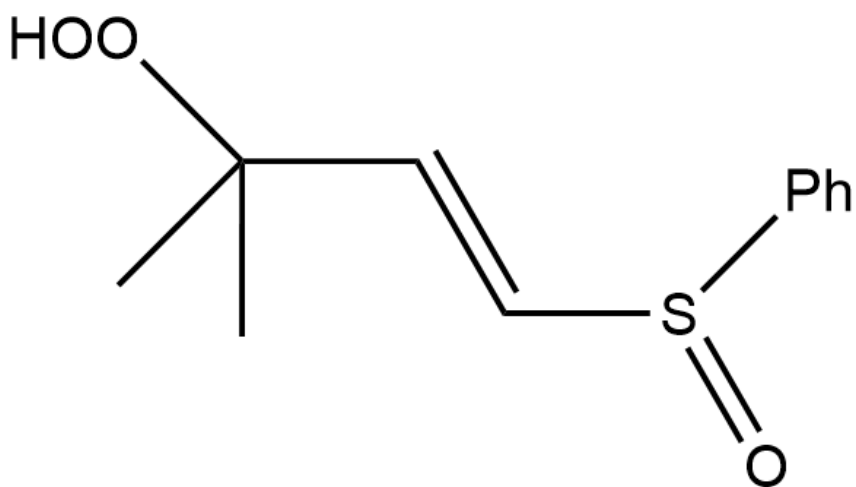
Product 2

D.



CC(C)(OO)/C=C/C(OC)=O, Product 1

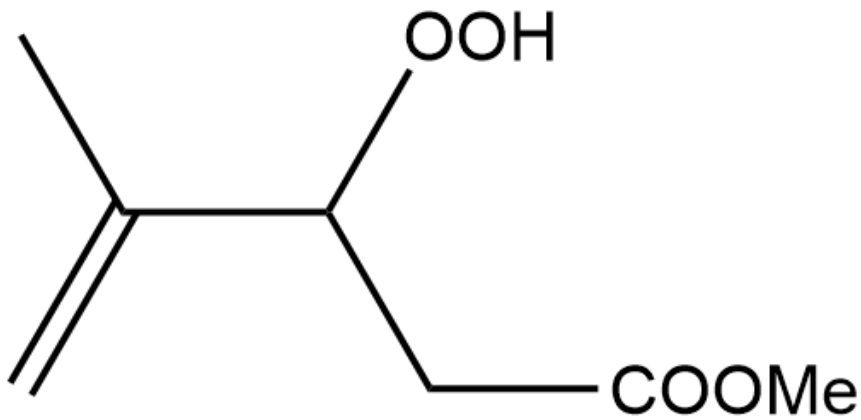
Product 1



O=S(C1=CC=CC=C1)/C=C/C(C)(OO)C, Product 2

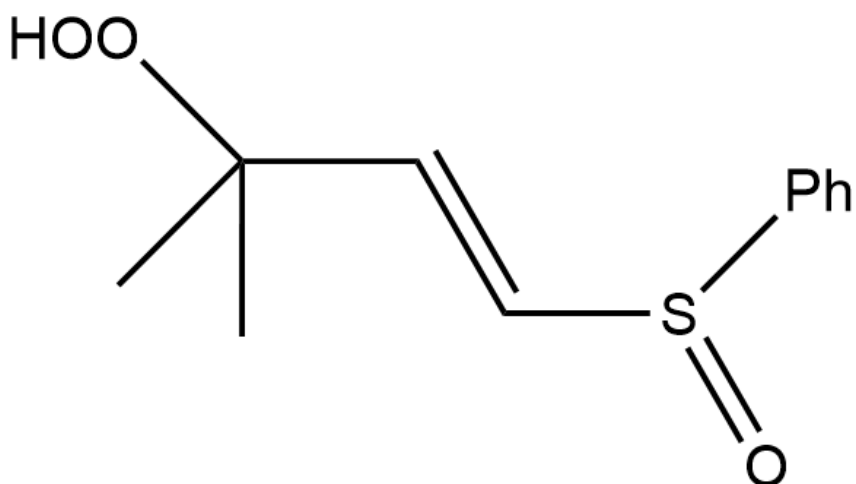
Product 2

E.



CC(C(OO)CC(OC)=O)=C, Product 1

Product 1



O=S(C1=CC=CC=C1)/C=C/C(C)(OO)C, Product 2

Product 2

Answer

Correct Answer: **B**

Detailed Explanation

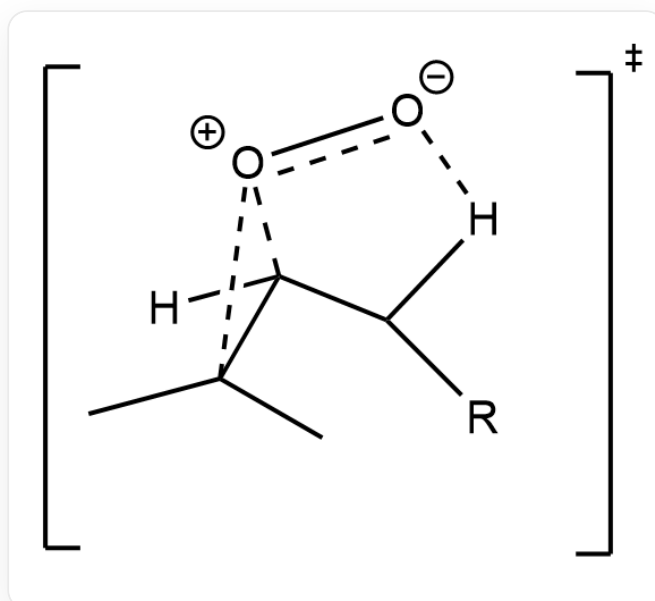
Compared to carbonyl groups, sulfoxides exhibit greater polarization with more concentrated negative charge on the oxygen.

CHECKPOINT

1 PTS

Compared to carbonyl groups, sulfoxides exhibit greater polarization with more concentrated negative charge on the oxygen

This is the transition state of the reaction (taking Reaction 1 as an example)



[H]C1(C([O+])1[O-])(C)C([R])[H]

For R=COOMe, attacking the hydrogen at the ortho-position of the carbonyl group, the carbonyl group can stabilize the transition state through conjugation, resulting in the conjugated product.

CHECKPOINT

1 PTS

For R=COOMe, attacking the hydrogen at the ortho-position of the carbonyl group, the carbonyl group can stabilize the transition state through conjugation, resulting in the conjugated product

For R=SOPh, the excessive negative charge on the oxygen creates strong repulsion between the peroxy group and sulfoxide oxygen in the transition state when attacking the ortho-position hydrogen, thus favoring the formation of terminal alkenes.

CHECKPOINT

1 PTS

For R=SOPh, the excessive negative charge on the oxygen creates strong repulsion between the peroxy group and sulfoxide oxygen in the transition state when attacking the ortho-position hydrogen, thus favoring the formation of terminal alkenes